

Solutions to First Problem Sheet

on

Linear Algebra

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Topic: Linear Spaces

“Mathematics is the art of reducing any problem to linear algebra.”
-William Stein

Problem 1

Prove the following statements.

- i) The identity element of a group is unique.
- ii) The inverse of a given element of a group is unique.
- iii) The inverse of a product is the product of the inverses in reversed order.
- iv) The cancellation law holds true: If Y, Z are elements of a given group such that

$$XY = XZ$$

for some element X in the group, then $Y = Z$. Similarly,

$$YX = ZX$$

implies the same conclusion.

- v) For any group G we have $fG = G$ for any $f \in G$ where $fG = \{fg : g \in G\}$.

Solution 1

Let G be a group. Then

- i) Suppose I, I' are two identities in G , then $II' = I$ because I' is an identity and $II' = I'$ because I is an identity too, thus $I = I'$.
- ii) Let $g \in G$, and suppose that g_1^{-1}, g_2^{-1} are two inverses for g , then $g_1^{-1}g = I$ because g_1^{-1} is an inverse for g , but then $g_1^{-1}gg_2^{-1} = g_2^{-1}$ and the LHS equals g_1^{-1} because g_2^{-1} is an inverse for g , so the conclusion now readily follows.
- iii) Let $g, h \in G$, then $gh(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = gg^{-1} = I$.
- iv) Suppose $Y, Z \in G$ are such that $XY = XZ$ for some $X \in G$, then since G is a group, X has an inverse and thus $X^{-1}XY = X^{-1}XZ$ from which the conclusion is clear. The other part is done in a similar fashion.
- v) Let $f \in G$, then for any given $g \in G$ we look for an element $t \in G$ such that $ft = g$, multiplying both sides on the left by the inverse of f we find that $t = f^{-1}g$, which is an element of G again, and thus the proof is done.



Problem 2

The definition of a linear space is usually written in terms of an explicit list of rules that a set equipped with binary operations must satisfy. Using our definition of a linear space (in terms of semigroups, groups, and bodies) over a body, state the complete explicit list of axioms that set must satisfy in order for it to

be a linear space. Such list is usually called the **linearity axioms**.

Solution 2

For a linear space $\mathcal{V}$ over a body $\mathbb{K}$, the full list of properties is given by

- i) $u + v \in \mathcal{V}$
- ii) $(u + v) + w = u + (v + w)$
- iii) $v + 0 = v$
- iv) $v + (-v) = 0$
- v) $v + w = w + v$
- vi) $\alpha v \in \mathcal{V}$
- vii) $\alpha(\beta v) = (\alpha\beta)v$
- viii) $(\alpha + \beta)v = \alpha v + \beta v$
- ix) $\alpha(v + w) = \alpha v + \alpha w$
- x) $1v = v$

for all $u, v, w \in \mathcal{V}$ and all $\alpha, \beta \in \mathbb{K}$, where 1 is the multiplicative identity of the body and 0 the additive identity of the Abelian group $\mathcal{V}$. To be more specific, i)–iv) correspond to the fact that $(V, +)$ is a group, v) to the group being Abelian, and vi)–x) to the definition of scalar multiplication given in the course. ☆

Problem 3

Show that bullets ii), iii), v), ix) in Example 1.5 are all indeed examples of linear spaces.

Hint: for bullet ix) you may appeal to the usual theorems for combinations of continuous function from complex analysis.

Solution 3

- ii) In this case, vector addition is understood as the addition of complex numbers, from complex analysis, we know that $(\mathbb{C}, +)$ is indeed an Abelian group, so take the body as $\mathbb{R}$ and recall that a real number times a complex number is another complex number, more generally, such a multiplication can be understood as multiplying a complex number (with imaginary part equal to zero) by another complex number so that multiplication by a scalar satisfies the required properties (vi)– to x)) in Problem 2.
- iii) The arguments are similar to the previous case, one can distinguish the complex numbers in $\mathcal{V}$ from those in the actual body by writing $\underline{z} \in \mathcal{V}$ and $z \in \mathbb{C}$ for any $z = x + iy$.
- v) The usual vector addition and multiplication by a scalar are defined componentwise, so the properties are straightforward.
- ix) From complex analysis, we know that the for any subset $\Omega \subseteq \mathbb{C}$, linear combinations of functions in $C^m(\Omega)$ yield a function in $C^m(\Omega)$ for any integer n and thus this is a linear space over $\mathbb{C}$.

☆

Problem 4

Let $\mathcal{V}$ be a linear space over $\mathbb{K}$. For $\alpha \in \mathbb{K}$ and $v \in \mathcal{V}$, show that $\alpha v = 0$ implies $\alpha = 0$ or $v = 0$.

Solution 4

For the sake of the cleanness of the proof, let us denote the zero vector by $\mathbf{0}$ and the additive identity of the body by 0. Let $\alpha v = \mathbf{0}$, then either $\alpha = 0$, in which case we have that $0v = (\lambda - \lambda)v = \lambda v - \lambda v = \mathbf{0}$, so the equation holds true. On the other hand, if $\alpha \neq 0$, we can multiply both sides by its multiplicative inverse to get $v = \alpha^{-1}\mathbf{0}$, now, notice that for any given scalar λ we have $\lambda\mathbf{0} = \lambda(u - u) = \lambda u - \lambda u = \mathbf{0}$,

so that in particular, $a^{-1}\mathbf{0} = \mathbf{0}$ and thus $v = a^{-1}\mathbf{0} = \mathbf{0}$ and hence $v = \mathbf{0}$. We therefore conclude that either $\alpha = 0$ or $v = \mathbf{0}$. ☆

Problem 5

Let ∞ and $-\infty$ be two distinct objects, neither of which is in $\mathbb{R}$. Define an addition and scalar multiplication on $\mathbb{R} \cup \{\infty, -\infty\}$ by

$$t\infty = \begin{cases} -\infty & t < 0, \\ 0 & t = 0, \\ \infty & t > 0, \end{cases} \quad t(-\infty) = \begin{cases} \infty & t < 0, \\ 0 & t = 0, \\ -\infty & t > 0, \end{cases}$$

and

$$\begin{aligned} t + \infty &= \infty + t = \infty + \infty = \infty, \\ t + (-\infty) &= (-\infty) + t = (-\infty) + (-\infty) = -\infty, \\ \infty + (-\infty) &= (-\infty) + \infty = 0. \end{aligned}$$

With these operations of addition and scalar multiplication, is $\mathbb{R} \cup \{-\infty, \infty\}$ a linear space over $\mathbb{R}$?

Note: These definitions (except the last one, where $-\infty + \infty = 0$, are actually used in Measure Theory to define the set of extended real numbers and work with the Lebesgue integral).

Solution 5

Yes ☆

Problem 6

Let $\beta = \{v_j\}_{j=1}^n$ and let $\beta_m = \{v_j\}_{j=1}^m$ for $m = 1, \dots, n$. Show that β is a linearly independent set if and only if $v_k \notin \text{span } \beta_{k-1}$ for all $k \geq 1$ (here, we define $\beta_0 = \emptyset$ and understand that $\text{span } \emptyset = \{0\}$).

Solution 6

($\Leftarrow$) Suppose that $v_k \notin \text{span } \beta_{k-1}$ for any $1 \leq k \leq n$. We proceed by proving the contrapositive. Assume that $\{v_j\}_{j=1}^n$ is a linearly dependent set, i.e., assume that there exists a set $\Sigma = \{\alpha_1, \alpha_2, \dots, \alpha_n\} \neq \{0\}$ such that $\alpha_j v_j = 0$, then take α_z be a nonzero element of Σ

$$\alpha_j v_j = \sum_{\substack{k \neq z \\ k=1}}^n \alpha_k v_k + \alpha_z v_z = 0 \implies v_z = -\frac{1}{\alpha_z} \sum_{\substack{k \neq z \\ k=1}}^n \alpha_k v_k$$

which on rebranding $-\alpha_k/\alpha_z = \zeta_k$ we get

$$v_z = \sum_{\substack{k \neq z \\ k=1}}^n \zeta_k v_k$$

so we can re-index the set $\{v_j\}_{j=1}^n$ in such way that $v_z = v_n$ and thus $v_z \in \text{span } \beta_{n-1}$.

($\Rightarrow$) Conversely, if $\{v_k\}_{k=1}^n$ is linearly independent, then

$$\alpha_j v_j = 0 \implies \alpha_j = 0 \quad \text{for all } j.$$

Suppose that there is some m such that $v_m \in \text{span } \beta_{m-1}$, then

$$v_m = \sum_{j=1}^{m-1} \gamma_j v_j \iff \sum_{j=1}^{m-1} \gamma_j v_j - v_m = 0$$

for some scalars $\{\gamma_1, \dots, \gamma_{m-1}\}$, and if we define $\gamma_m = -1$, then $\gamma_j v_j = 0$ ($j = 1, \dots, m$) but this implies that $\gamma_j = 0$ for all j , in particular, for $j = m$ we would get $\gamma_m = 0$, clearly, a contradiction, and the proof is complete. $\star$

Problem 7

For each of the following subsets of $\mathbb{C}^3$, determine whether it is a subspace of $\mathbb{C}^3$.

- i) $\{(\alpha_1, \alpha_2, \alpha_3) : \alpha_1 + 2\alpha_2 + 3\alpha_3 = 0\}$
- ii) $\{(\alpha_1, \alpha_2, \alpha_3) : \alpha_1 + 2\alpha_2 + 3\alpha_3 = 4\}$
- iii) $\{(\alpha_1, \alpha_2, \alpha_3) : \alpha_1 \alpha_2 \alpha_3 = 0\}$
- iv) $\{(\alpha_1, \alpha_2, \alpha_3) : \alpha_1 = 5\alpha_3\}$

Moreover, identify (geometrically) the subspaces in i) and iv).

Solution 7

- i) Yes ii) No iii) No iv) Yes
- i) and ii) are planes in $\mathbb{R}^3$.



Problem 8

Show that the subset S of $\mathcal{M}^{2 \times 2}$ of all symmetric matrices is a linear subspace of $\mathcal{M}^{2 \times 2}$. Find a basis for S and thus show that this subspace is three-dimensional.

Solution 8

The fact that is a linear subspace is obvious. For a basis we choose a slight variation of the canonical basis for $\mathcal{M}^{2 \times 2}$, namely,

$$B = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}$$

so that the subspace is three-dimensional. $\star$

Problem 9

Show that the intersection of two subspaces of a linear space is a subspace of that same linear space.

Solution 9

Let $\mathcal{U}, \mathcal{W}$ be subspaces of $\mathcal{V}$, then for any $|f\rangle, |g\rangle \in \mathcal{U} \cap \mathcal{W}$ one has that $|f\rangle + \lambda|g\rangle$ belongs to $\mathcal{U}$ since $\mathcal{U}$ is a linear subspace and similarly for $\mathcal{W}$ so that the linear combination is still in $\mathcal{U} \cap \mathcal{W}$ and the proof is done. $\star$

Problem 10

Prove Theorem 1.8, 1.9, and 1.11.

Solution 10

Proof 1.8) Let $\{v^k\}_{k=1}^N, \{v'^k\}_{k=1}^N$ be the components of a basis $\{|e_k\rangle\}_{k=1}^N$ of $|v\rangle$, then

$$|v\rangle = |e_k\rangle v^k = |e_k\rangle v'^k \iff (v^k - v'^k)|e_k\rangle = 0$$

but since the basis is a linearly independent set, we conclude that $v^k = v'^k$ for all k .

Proof 1.11) Given any inner product g , notice that

$$g(v, 0) = g(v, w - w) = g(v, w) - g(v, w) = 0$$

so that $g(v, 0) = 0$ and thus $g(v, 0) = g(0, v) = 0$. For the other statement, we simply use sesquisymmetry to see that

$$g(\alpha u + v, w) = g(w, \alpha u + v)^* = (\alpha g(w, u) + g(w, v))^* = \alpha^* g(w, u)^* + g(w, v)^* = \alpha^* g(u, w) + g(v, w).$$

This property is why (in this context) linearity in the second slot is called sesquilinearity. $\star$

Problem 11

Given a basis $\{a_i\}_{i=1}^N$ for a linear space $\mathcal{V}$, construct an orthonormal basis by taking linear combinations of the original one. More specifically, find a basis $\{e_i\}_{i=1}^N$ such that $\|e_i\| = 1$ for all i , such that $g(e_i, e_j) = \delta_{ij}$ and $e_i = \alpha_j a_j$ for some set $\{\alpha_j\}_{j=1}^N$ of scalars for each element e_i . This procedure is usually called the **Gram-Schmidt process**.

Solution 11

Consider a basis $B = \{|a\rangle_1, \dots, |a\rangle_N\}$ of a linear space $\mathcal{V}$, then we define $|e_1\rangle = |a_1\rangle / \|a_1\|$. Next, subtract from $|a_1\rangle$ from its projection along $|e_1\rangle$ and call the resultant vector $|e'_2\rangle$, this way, $|e'_2\rangle = |a_2\rangle - |e_1\rangle \langle e_1 | a_2 \rangle$ so that $\langle e_1 | e'_2 \rangle = 0$ and now we normalise $|e'_2\rangle$ and call the resultant vector $|e_2\rangle$. This way, after having calculated m orthonormal vectors $|e_1\rangle, \dots, |e_m\rangle$ ($m \leq N$), we can compute the next one by using

$$|e_{m+1}\rangle = \frac{1}{\|e'_{m+1}\|} |e'_{m+1}\rangle, \quad |e'_{m+1}\rangle = |a_{m+1}\rangle - \sum_{i=1}^m |e_i\rangle \langle e_i | a_{m+1} \rangle.$$

$\star$

Problem 12

Show that for any set of n complex numbers $\{\alpha_j\}_{j=1}^n$ we have

$$\left| \sum_{i=1}^n \alpha_i \right|^2 \leq n \sum_{i=1}^n |\alpha_i|^2.$$

Hint: Apply the Schwarz inequality to $(\alpha_1, \alpha_2, \dots, \alpha_n)$ and another appropriate vector.

Solution 12

Let $|1\rangle = (1, 1, \dots)$ and thus by the Cauchy-Schwarz inequality we have

$$|\langle 1 | \alpha \rangle|^2 = \left| \sum_{i=1}^n \alpha_i \right|^2 \leq n \sum_{i=1}^n |\alpha_i|^2 = \langle \alpha | \alpha \rangle \langle 1 | 1 \rangle$$

$\star$

Problem 13

Verify that $\mathbf{C} \in \text{End}(\mathcal{V})$ where $\mathcal{V}$ is $\mathbb{C}$ over $\mathbb{R}$ and

$$\mathbf{C}|z\rangle = |z^*\rangle.$$

Is $\mathbf{C}$ linear if instead of real numbers, complex numbers are used as scalars?

Solution 13

Since the scalars are real, we notice that

$$\mathbf{C}(|z\rangle + b|w\rangle) = |z^*\rangle + b|w^*\rangle.$$

If instead of real numbers, complex numbers are used as scalars, then we would get

$$\mathbf{C}(|z\rangle + \lambda|w\rangle) = |z^*\rangle + \lambda^*|w^*\rangle,$$

so that $\mathbf{C}$ would not be linear. ✧

Problem 14

Define the **kernel** of $\mathbf{T} \in \text{Hom}(\mathcal{V}, \mathcal{W})$ as

$$\ker \mathbf{T} = \{|v\rangle \in \mathcal{V} : \mathbf{T}|v\rangle = |0\rangle\},$$

and hence prove that $\ker \mathbf{T}$ is a subspace of $\mathcal{V}$ and that $\mathbf{T}(\mathcal{V})$ is a subspace of $\mathcal{W}$.

Solution 14

Let $|v\rangle, |v'\rangle \in \ker \mathbf{T}$. Then

$$\mathbf{T}(|v\rangle + \lambda|v'\rangle) = \mathbf{T}(|v\rangle) + \lambda\mathbf{T}(|v'\rangle) = |0\rangle + \lambda|0\rangle = |0\rangle.$$

Similarly, for $\mathbf{T}(\mathcal{V})$ we let $|w\rangle, |w'\rangle \in \mathbf{T}(\mathcal{V})$, then

$$|w\rangle + \lambda|w'\rangle = \mathbf{T}(|v\rangle) + \lambda\mathbf{T}(|v'\rangle) = \mathbf{T}(|v\rangle + \lambda|v'\rangle)$$

since $|v\rangle + \lambda|v'\rangle \in \mathcal{V}$ provided that $|v\rangle, |v'\rangle \in \mathcal{V}$, the conclusion now readily follows. ✧

Problem 15

Show that the map $d : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$ defined as in Theorem 1.29 is indeed a metric.

Solution 15

The definition is $d(u, v) = \|u - v\|$ from which is clear that $d(u, v) \neq 0$, $d(u, v) = 0$ if and only if $u = v$ and also that $d(u, v) \leq d(u, w) + d(w, v)$. ✧

Problem 16

Show that the map $\|\cdot\| : \mathcal{V} \rightarrow \mathbb{R}$ defined as in Theorem 1.31 is indeed a norm.

Solution 16

The homogeneity and positive definiteness property are readily seen, so we only prove the triangle inequality. For this, notice how

$$||u + v||^2 = \langle u|u \rangle + \langle v|v \rangle + 2\operatorname{Re}(\langle u|v \rangle) \leq \langle u|u \rangle + \langle v|v \rangle + 2|\langle u|v \rangle|$$

so that by the Cauchy-Bunyakovski-Schwarz inequality, one has $||u + v||^2 \leq (||u|| + ||v||)^2$ from which the conclusion now readily follows after applying the positive definiteness of the norm and taking the square root of both sides. $\star$

Problem 17

Show that if $(\alpha_n), (\beta_n)$ are in ℓ^2 , then

$$\sum_{j \in \mathbb{N}} \alpha_j^* \beta_j$$

is convergent.

Solution 17

Notice that the term is just the inner product $\langle \alpha | \beta \rangle$ so that by the Schwarz inequality one has

$$|\langle \alpha | \beta \rangle| \leq ||\alpha|| ||\beta|| = \sqrt{\sum |\alpha_i|^2} \sqrt{\sum |\beta_j|^2}$$

so given that $\alpha, \beta \in \ell^2$, the conclusion $\sum \alpha_n^* \beta_n < \infty$ now readily follows. $\star$

Problem 18

Express the following permutations as products of disjoint cycles, and determine which are cyclic.

- i) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 3 & 4 & 5 & 6 & 2 \end{pmatrix}$
- ii) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 4 & 5 & 6 & 3 \end{pmatrix}$
- iii) $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 5 & 4 & 2 \end{pmatrix}$

Solution 18

- i) $(1)(2\ 3\ 4\ 5\ 6)$ Cyclic of length 5
- ii) $(1\ 2)(3\ 4\ 5\ 6)$ Not cyclic
- iii) $(1)(2\ 3\ 5)(4)$ Cyclic of length 3

$\star$

Problem 19

Show that if $\pi \in S_n$, then π and π^{-1} have the same parity.

Solution 19

Given any permutation π , the definition of π^{-1} implies that $\pi^{-1}\pi = e$ but since e is an even permutation, we require that π and π^{-1} have the same parity (product of permutations with distinct parities yield an odd permutation). $\star$

Problem 20

Prove that the determinant of a linear operator $\mathbf{A}$ has the following properties:

- i) If $\mathbf{A} = \lambda \mathbf{1}$, then $\det \mathbf{A} = \lambda^N$.
- ii) $\mathbf{A}$ is invertible if and only if $\det \mathbf{A} \neq 0$.
- iii) $\det(\mathbf{A} \circ \mathbf{B}) = \det \mathbf{A} \det \mathbf{B}$.

Solution 20

Recall that $\det \mathbf{A}$ is defined via $\Delta_A(|e_1\rangle, \dots, |e_N\rangle) = \det \mathbf{A} \cdot \Delta(|e_1\rangle, \dots, |e_N\rangle)$ with Δ a nonzero determinant function and $\{|e_i\rangle\}_{i=1}^N$ a basis for $\mathcal{V}$. Then

- i) If $\mathbf{A} = \lambda \mathbf{1}$, then

$$\Delta_A(|e_1\rangle, \dots, |e_N\rangle) = \Delta(\lambda \mathbf{1}|e_1\rangle, \dots, \lambda \mathbf{1}|e_N\rangle) = \lambda^N \Delta(|e_1\rangle, \dots, |e_N\rangle)$$

so that $\Delta_A = \lambda^N \Delta$.

- ii) ($\Rightarrow$) Suppose $\mathbf{A}$ is invertible, this equivalent to say that $\mathbf{A}$ sends linearly independent sets onto linearly independent sets, but if $\mathbf{A}$ is invertible, then so is $\mathbf{A}^{-1}$. Since the definition of determinant is independent of the determinant function and independent of the basis, let $\{|e_i\rangle\}_{i=1}^N$ be any basis and thus $\{\mathbf{A}^{-1}|e_i\rangle\}_{i=1}^N$ is also a basis, now notice that

$$\begin{aligned} \Delta_A(\mathbf{A}^{-1}|e_1\rangle, \dots, \mathbf{A}^{-1}|e_N\rangle) &= \det \mathbf{A} \Delta(\mathbf{A}^{-1}|e_1\rangle, \dots, \mathbf{A}^{-1}|e_N\rangle) = \det \mathbf{A} \Delta_{\mathbf{A}^{-1}}(|e_1\rangle, \dots, |e_N\rangle) \\ &= \det \mathbf{A} \det \mathbf{A}^{-1} \Delta(|e_1\rangle, \dots, |e_N\rangle), \end{aligned}$$

where in the last equality I used the definition of determinant for $\mathbf{A}^{-1}$. It is now readily seen that

$$\Delta(|e_1\rangle, \dots, |e_N\rangle) = \det \mathbf{A} \det \mathbf{A}^{-1} \Delta(|e_1\rangle, \dots, |e_N\rangle),$$

and since Δ is a nonzero determinant function, I can divide both sides by $\Delta(|e_1\rangle, \dots, |e_N\rangle)$ to get

$$1 = \det \mathbf{A} \det \mathbf{A}^{-1},$$

therefore $\det \mathbf{A} \neq 0 \neq \det \mathbf{A}^{-1}$. Furthermore, if the latter holds, then

$$\det \mathbf{A}^{-1} = \frac{1}{\det \mathbf{A}}.$$

Which is a standard result in a first course on linear algebra.

($\Leftarrow$) I will show the converse using the contrapositive. Assume that $\mathbf{A}$ is not invertible and choose a basis $\{|e_j^*\rangle\}_{j=1}^N$ with the property that $\{\mathbf{A}|e_j^*\rangle\}_{j=1}^N$ is a linearly dependent set (if no such basis exists, then $\mathbf{A}$ would send all linearly independent sets onto linearly independent sets and that would imply that $\mathbf{A}$ is invertible). The fact that $\{\mathbf{A}|e_j^*\rangle\}_{j=1}^N$ is linearly dependent together and the fact that Δ is a determinant function imply that $\Delta(\mathbf{A}|e_1^*\rangle, \dots, \mathbf{A}|e_N^*\rangle) = 0$ but by the definition of determinant (choosing the starred basis)

$$0 = \Delta(\mathbf{A}|e_1^*\rangle, \dots, \mathbf{A}|e_N^*\rangle) = \det \mathbf{A} \cdot \Delta(|e_1^*\rangle, \dots, |e_N^*\rangle),$$

such equation can only be satisfied if $\det \mathbf{A} = 0$ because $\Delta(|e_1^*\rangle, \dots, |e_N^*\rangle) \neq 0$.

- iii) For the sake of economy of notation we will write $\Delta(|e_1\rangle, \dots, |e_N\rangle) \equiv \Delta\{|e_j\rangle\}$. It can be readily seen that

$$\Delta_{\mathbf{A} \circ \mathbf{B}}\{|e_j\rangle\} = \Delta_{\mathbf{A}}\{\mathbf{B}|e_j\rangle\} = \det \mathbf{A} \cdot \Delta\{\mathbf{B}|e_j\rangle\} = \det \mathbf{A} \det \mathbf{B} \cdot \Delta\{|e_j\rangle\},$$

from which is clear that $\Delta_{\mathbf{A} \circ \mathbf{B}} = \det \mathbf{A} \det \mathbf{B} \cdot \Delta$ and the result is now obvious from the definition of determinant. $\star$

Problem 21

Let Δ be a determinant function in the N -dimensional linear space $\mathcal{V}$. Let $|v\rangle$ and $\{|v_k\rangle\}_{k=1}^N$ be vectors in $\mathcal{V}$. Then show that

$$\sum_{j=1}^N (-1)^{j-1} \Delta \left(|v\rangle, |v_1\rangle, \dots, \widehat{|v_j\rangle}, \dots, |v_N\rangle \right) |v_j\rangle = \Delta(|v_1\rangle, \dots, |v_N\rangle) |v\rangle$$

where a hat on a vector means that particular vector is missing.

Solution 21

We wish to establish the equation

$$\sum_{j=1}^N (-1)^{j-1} \Delta \left(|v\rangle, |v_1\rangle, \dots, \widehat{|v_j\rangle}, \dots, |v_N\rangle \right) |v_j\rangle = \Delta(|v_1\rangle, \dots, |v_N\rangle) |v\rangle \quad (\dagger)$$

First, suppose that $\{|v_k\rangle\}_{k=1}^N$ is a linearly dependent set, then there exist numbers c_k not all zero such that $c_k |v_k\rangle = 0, k = 1, 2, \dots, N$. In particular, take some $c_m \neq 0$, then we can conclude that

$$|v_m\rangle = -\frac{1}{c_m} \sum_{\substack{k \neq m \\ k=1}}^N c_k |v_k\rangle. \quad (*)$$

Let us now define

$$\phi_j(|v\rangle) = (-1)^{j-1} \Delta \left(|v\rangle, |v_1\rangle, \dots, \widehat{|v_j\rangle}, \dots, |v_N\rangle \right).$$

We claim that for all j , the relation $c_m \phi_j(|v\rangle) = c_j \phi_m(|v\rangle)$ holds. To prove this, we first note that the case $j = m$ is trivially true and thus consider $j \neq m$ so that using $(*)$ and linearity of Δ we get

$$\begin{aligned} \phi_j(|v\rangle) &= (-1)^{j-1} \Delta \left(|v\rangle, |v_1\rangle, \dots, \widehat{|v_j\rangle}, \dots, |v_m\rangle, \dots, |v_N\rangle \right) \\ &\quad - \frac{1}{c_m} \sum_{\substack{k \neq m \\ k=1}}^N (-1)^{j-1} c_k \Delta \left(|v\rangle, |v_1\rangle, \dots, \widehat{|v_j\rangle}, \dots, \underbrace{|v_k\rangle}_{m\text{th slot}}, \dots, |v_N\rangle \right), \end{aligned}$$

here, notice that the determinant function evaluates to 0 when $k \neq j$ since the entry is then repeated (see **Problem 3**), so the only term surviving in the sum is the one for which $k = j$, i.e.,

$$\phi_j(|v\rangle) = -\frac{1}{c_m} (-1)^{j-1} c_j \Delta \left(|v\rangle, |v_1\rangle, \dots, \widehat{|v_j\rangle}, \dots, \underbrace{|v_j\rangle}_{m\text{th slot}}, \dots, |v_N\rangle \right),$$

moving the element $|v_j\rangle$ $|m-j|$ times to the left or to the right, depending on whether it was originally to the right or to the left, respectively, we have that such a movement adds a factor of $(-1)^{|m-j|}$ and now the m th slot is the one with the "missing" vector so

$$\phi_j(|v\rangle) = -\frac{1}{c_m} (-1)^{j-1} (-1)^{|m-j|} c_j \Delta \left(|v\rangle, |v_1\rangle, \dots, |v_j\rangle, \dots, \widehat{|v_m\rangle}, \dots, |v_N\rangle \right),$$

and since $(-1)^r = (-1)^{-r}$ we can drop the absolute value in the sign factor to get

$$\begin{aligned} \phi_j(|v\rangle) &= -\frac{1}{c_m} (-1)^{j-1} (-1)^{m-j} c_j \Delta \left(|v\rangle, |v_1\rangle, \dots, \widehat{|v_m\rangle}, \dots, |v_N\rangle \right) \\ &= -\frac{1}{c_m} (-1)^{m-1} c_j \Delta \left(|v\rangle, |v_1\rangle, \dots, \widehat{|v_m\rangle}, \dots, |v_N\rangle \right), \end{aligned}$$

so that $\phi_j(|v\rangle) = \frac{c_j}{c_m} \phi_m(|v\rangle)$ (recall that $c_m \neq 0$), as claimed. Now we notice that given that $c_j |v_j\rangle = 0$,

$$\phi_j(|v\rangle) |v_j\rangle = \frac{c_j}{c_m} \phi_m(|v\rangle) |v_j\rangle = \frac{\phi_m(|v\rangle)}{c_m} c_j |v_j\rangle = 0 \quad (\text{no sum on } m),$$

and hence, for $\{|v_k\rangle\}_{k=1}^N$ linearly dependent,

$$\sum_{j=1}^N (-1)^{j-1} \Delta(|v\rangle, |v_1\rangle, \dots, \widehat{|v_j\rangle}, \dots, |v_N\rangle) |v_j\rangle = 0,$$

and as an immediate consequence of the fact that $\Delta \in \Lambda^N(\mathcal{V})$ we get that $\Delta(|v_1\rangle, \dots, |v_N\rangle) = 0$ because the elements in the argument are linearly dependent. We therefore conclude that $(\dagger)$ holds. Now, for the other case, suppose that $\{|v_k\rangle\}_{k=1}^N$ is a linearly independent set, then it is a basis for $\mathcal{V}$, and we can thus express $|v\rangle$ as $|v\rangle = \alpha_i |v_i\rangle$, in this case the determinant function on the LHS of $(\dagger)$ becomes

$$\alpha_i \Delta(|v_i\rangle, |v_1\rangle, \dots, \widehat{|v_j\rangle}, \dots, |v_N\rangle) \quad i = 1, 2, \dots, N,$$

for each j , but this evaluates to zero if $i \neq j$, again, because the entry $|v_i\rangle$ is repeated in the argument, and hence the only remaining terms in the sums are the ones with $j = i$, i.e., the LHS of $(\dagger)$ becomes

$$\sum_{i=1}^N (-1)^{i-1} \alpha_i \Delta(|v_i\rangle, |v_1\rangle, \dots, \widehat{|v_i\rangle}, \dots, |v_N\rangle) |v_i\rangle,$$

which on moving the first element in the entry $(|v_j\rangle)$ $i - 1$ times to the right we get

$$\sum_{i=1}^N (-1)^{i-1} (-1)^{i-1} \alpha_i \Delta(|v_1\rangle, \dots, |v_i\rangle, \dots, |v_N\rangle) |v_i\rangle = \sum_{i=1}^N \alpha_i \Delta(|v_1\rangle, \dots, |v_N\rangle) |v_i\rangle = \Delta(|v_1\rangle, \dots, |v_N\rangle) |v\rangle.$$

The conclusion now readily follows. ✧

Problem 22

Show that the classical adjoint satisfies the following relations:

$$\text{ad}(\mathbf{A}) \circ \mathbf{A} = \det \mathbf{A} \cdot \mathbf{1} = \mathbf{A} \circ \text{ad}(\mathbf{A}).$$

Where $\mathbf{1}$ is the identity operator.

Solution 22

Since $\text{ad}(\mathbf{A})$ is the unique operator in $\mathcal{V}$ satisfying

$$\sum_{j=1}^N (-1)^{j-1} \Delta(|v\rangle, \mathbf{A}|v_1\rangle, \dots, \widehat{\mathbf{A}|v_j\rangle}, \dots, \mathbf{A}|v_N\rangle) |v_j\rangle = \Delta(|v_1\rangle, \dots, |v_N\rangle) \text{ad}(\mathbf{A})|v\rangle,$$

we may replace $|v\rangle$ for $\mathbf{A}|v\rangle$ (the equation above must hold for any $|v\rangle \in \mathcal{V}$) to get

$$\sum_{j=1}^N (-1)^{j-1} \Delta(\mathbf{A}|v\rangle, \mathbf{A}|v_1\rangle, \dots, \widehat{\mathbf{A}|v_j\rangle}, \dots, \mathbf{A}|v_N\rangle) |v_j\rangle = \Delta(|v_1\rangle, \dots, |v_N\rangle) \text{ad}(\mathbf{A}) \circ \mathbf{A}|v\rangle.$$

Then the LHS can be written as

$$\det \mathbf{A} \cdot \sum_{j=1}^N (-1)^{j-1} \Delta \left(|v\rangle, |v_1\rangle, \dots, \widehat{|v_j\rangle}, \dots, |v_N\rangle \right) |v_j\rangle = \det \mathbf{A} \cdot \Delta(|v_1\rangle, \dots, |v_N\rangle) |v\rangle,$$

where the last equality follows from Theorem 1.50, and now using $|v\rangle = \mathbf{1}|v\rangle$ we see that the two previous equations combined yield

$$\Delta(|v_1\rangle, \dots, |v_N\rangle) \text{ad}(\mathbf{A}) \circ \mathbf{A}|v\rangle = \det \mathbf{A} \cdot \Delta(|v_1\rangle, \dots, |v_N\rangle) \mathbf{1}|v\rangle,$$

and since Δ is a nonzero determinant function it is readily seen that $\text{ad}(\mathbf{A}) \circ \mathbf{A} = \det \mathbf{A} \cdot \mathbf{1}$. For the second equality we apply $\mathbf{A}$ to the defining equation of $\text{ad}(\mathbf{A})$ to get

$$\sum_{j=1}^N (-1)^{j-1} \Delta \left(|v\rangle, \mathbf{A}|v_1\rangle, \dots, \widehat{\mathbf{A}|v_j\rangle}, \dots, \mathbf{A}|v_N\rangle \right) \mathbf{A}|v_j\rangle = \Delta(|v_1\rangle, \dots, |v_N\rangle) \mathbf{A} \circ \text{ad}(\mathbf{A})|v\rangle,$$

but by Theorem 1.50 the LHS becomes

$$\Delta(\mathbf{A}|v_1\rangle, \dots, \mathbf{A}|v_N\rangle) |v\rangle = \det \mathbf{A} \cdot \Delta(|v_1\rangle, \dots, |v_N\rangle) |v\rangle,$$

so it is readily seen that $\det \mathbf{A} \cdot \mathbf{1} = \mathbf{A} \circ \text{ad}(\mathbf{A})$.

